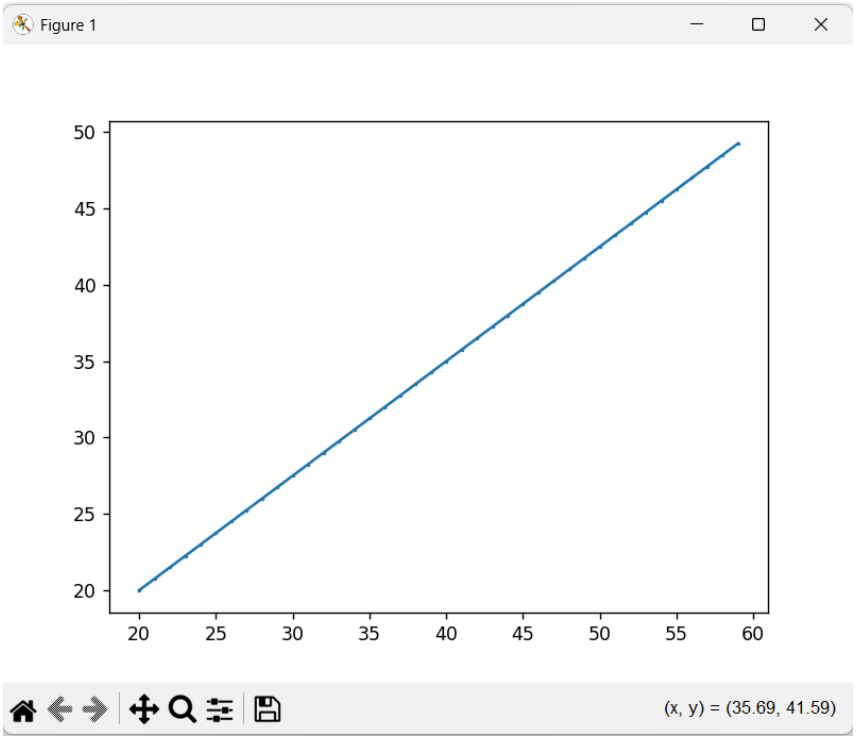
Figure 1
50
45
40
35
30
25
20
20
25
30
35
40
45
50
55
60
(x, y) = (35.69, 41.59)

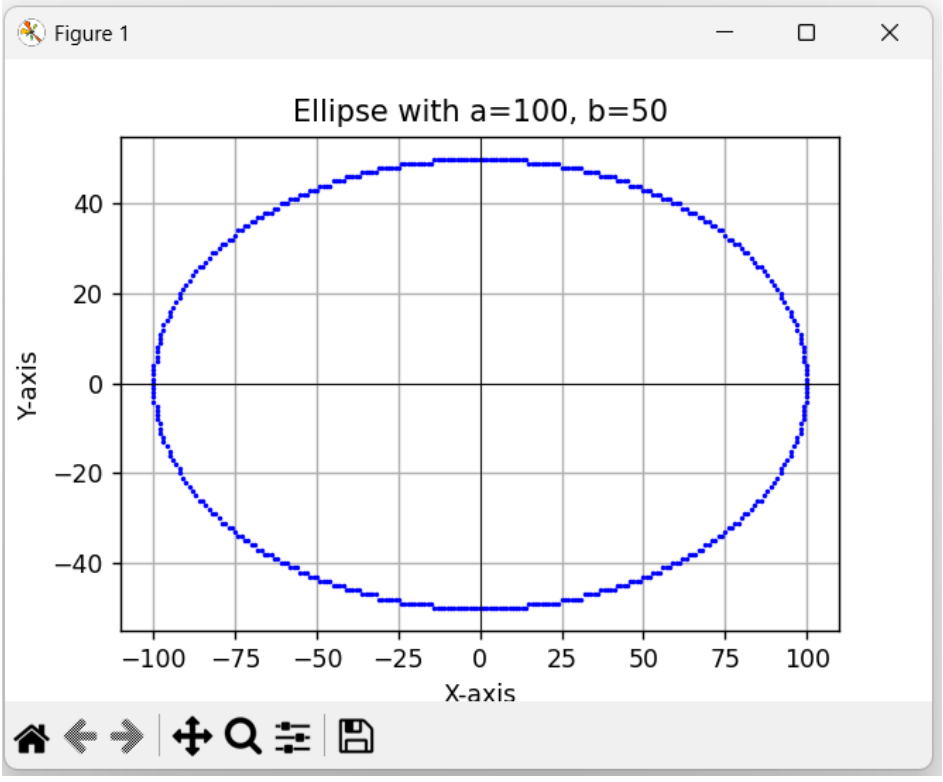
Figure 1
Ellipse with a=100, b=50
40
20
0
−20
−40
Y-axis
−100
−75
−50
−25
0
25
50
75
100
X-axis

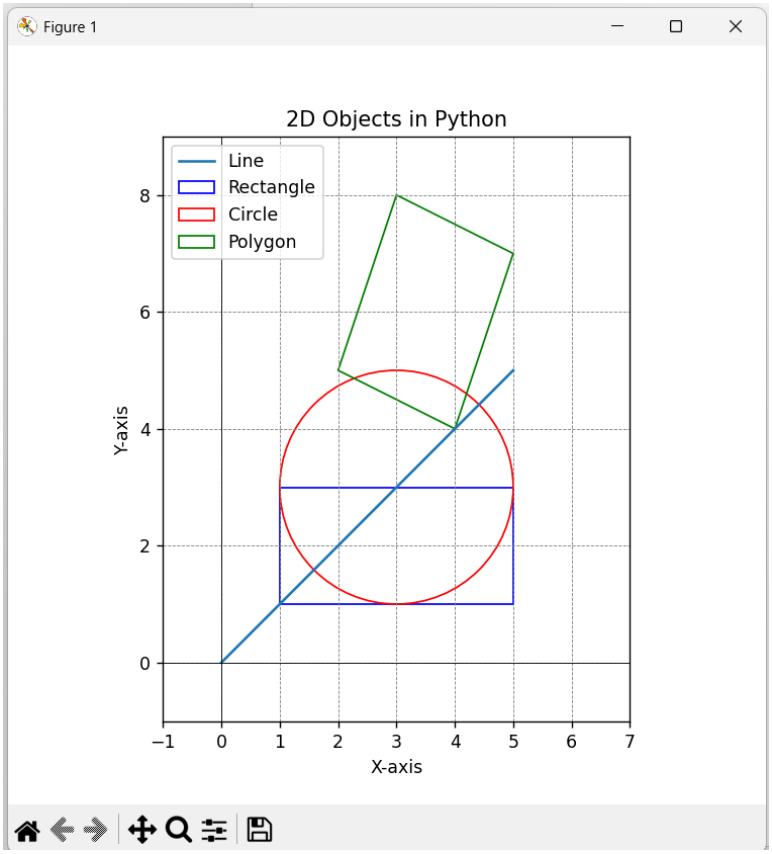
Figure 1
2D Objects in Python
Line
Rectangle
Circle
Polygon
Y-axis
X-axis
8
6
4
2
0
−1
0
1
2
3
4
5
6
7

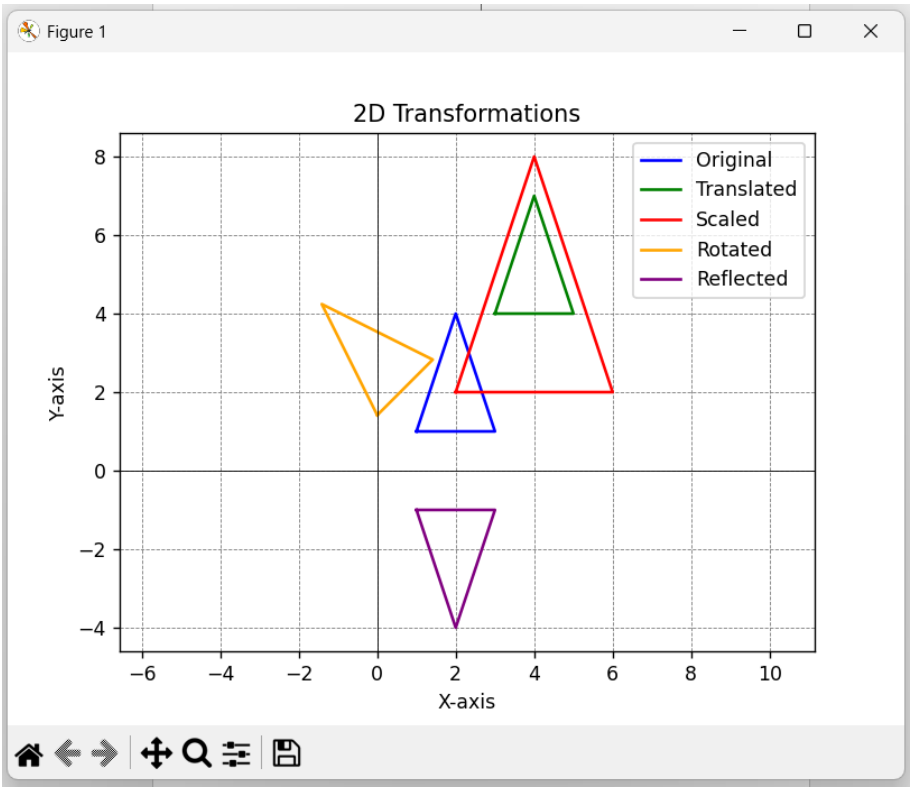
Figure 1
2D Transformations
Original
Translated
Scaled
Rotated
Reflected
Y-axis
X-axis
8
6
4
2
0
−2
−4
−6
−4
−2
0
2
4
6
8
10

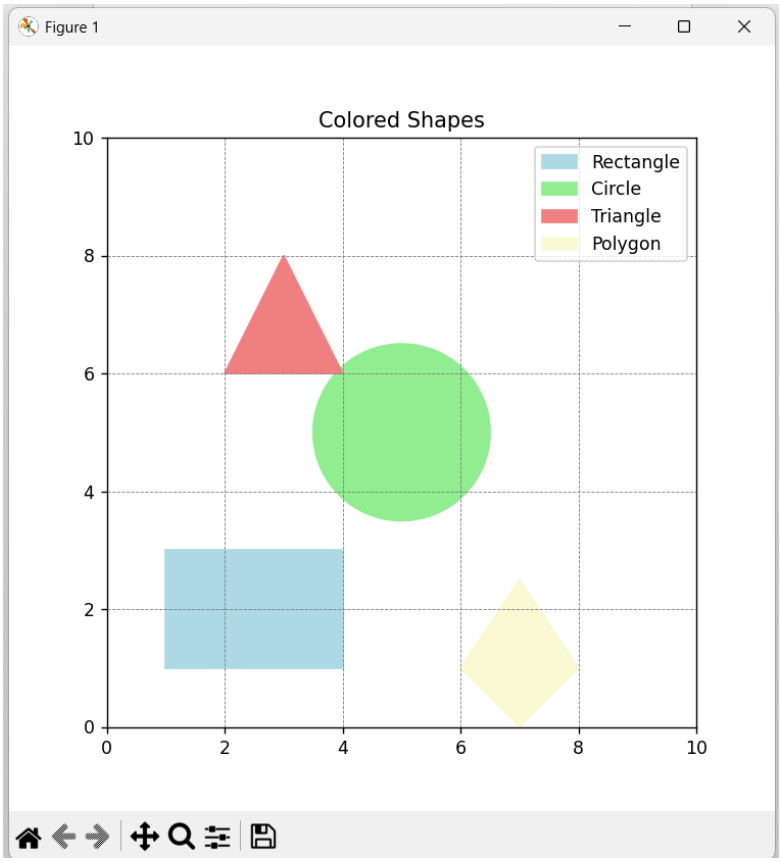
Figure 1
Colored Shapes
Rectangle
Circle
Triangle
Polygon
0
2
4
6
8
10

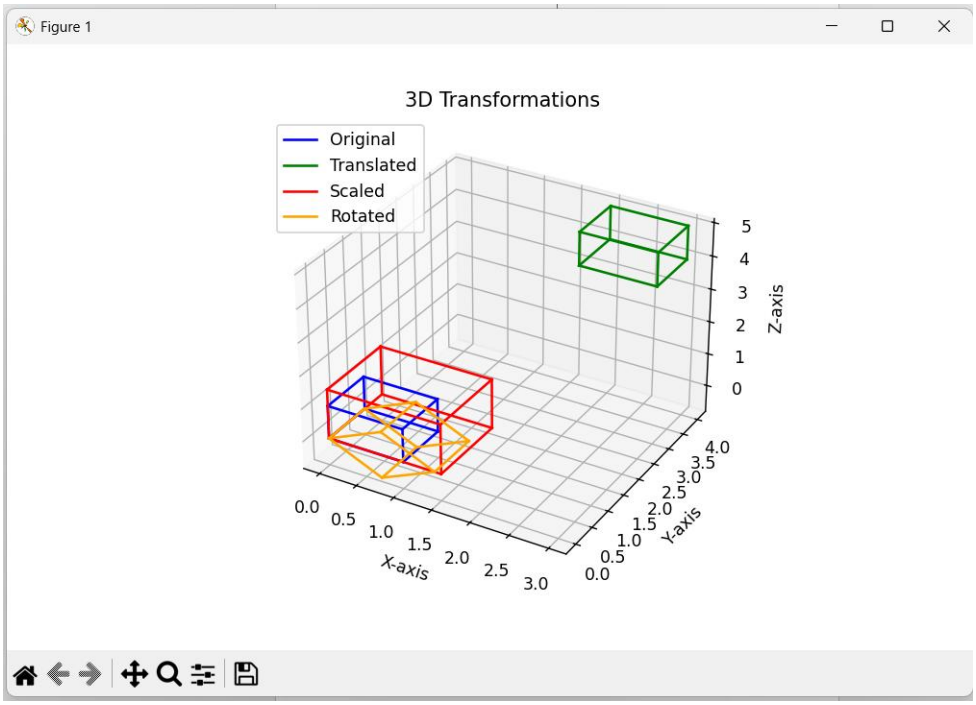
Figure 1
3D Transformations
Original
Translated
Scaled
Rotated
X-axis
Y-axis
Z-axis

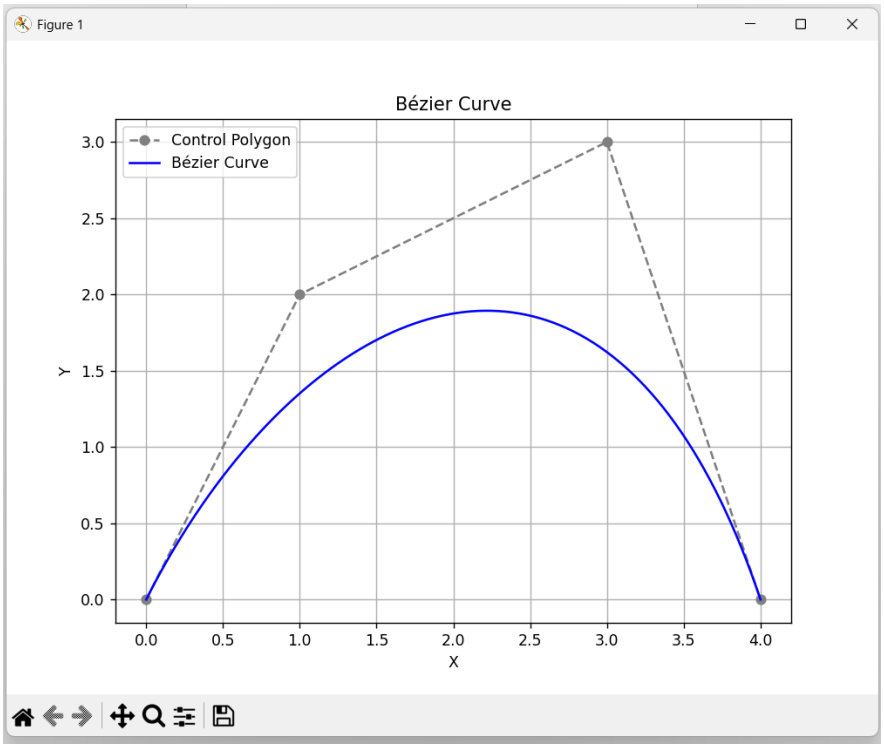
Figure 1
Bézier Curve
Control Polygon
Bézier Curve
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Y
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
X